GG-P-118C
December 11, 1975
SUPERSEDING
Fed. Spec. GG-P-118B
November 13, 1963

FEDERAL SPECIFICATION

PARALLEL RULERS, FOLDING TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers parallel rulers with wood blades or plastic blades of the stepping or folding type.

1.2 Classification. The rulers shall be of the following classes and sizes and identified by a type designation number.

1.2.1 Type designation number. Rulers covered by this specification shall be identified by a type designation number formed as indicated below. This type designation number is intended for cataloging and ordering purposes and not for surface printing on the rulers (see 6.2).

GG-P-118	1	06
Fed. Spec. No. (1.2.2)	Class of Ruler (1.2.3)	Size of Ruler (1.2.4)

1.2.2 Ruler Specification No. - GG-P-118.

1.2.3 Class of rulers shall be as indicated below:

1 - Wood blades.
2 - Plastic blades.

1.2.4 Size of ruler shall be as indicated below:

06 - 6 inch.
12 - 12 inch.
15 - 15 inch.
18 - 18 inch.
24 - 24 inch.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-398 - Plastic Molding Material, Styrene-Butadiene.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-665 - Boxes: Paperboard, Metal Edged and Components.
PPP-B-676 - Boxes, Set-Up.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Std. No. 356 - Commercial Packaging of Supplies and Equipment.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-M-20693 - Molding Plastic, Polyamide (Nylon), Rigid.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-1363 - Measurement of Wood Moisture Content.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The parallel ruler shall consist of two parallel blades linked together by two or three bars. The blades shall have knobs to facilitate spreading (see Figure 1).

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.2.1 Metal. All metal parts shall be of chrome-plated brass, anodized aluminum, or corrosion-resisting steel.

3.3 Blades. The blades shall be 3/16 inch, plus or minus 3/64 inch, thick. The length and width shall be as shown in table I. A tolerance of plus or minus 1/16 inch shall be permitted.

TABLE I. Blade Dimension

Size (inches)	Width (inches)
6	1
12	1-1/4
15	1-1/4
18	1-1/4
24	1-1/2

3.3.1 Edges. The ruling edge of each blade shall be beveled between 3/16 and 1/4 inch with a working surface between 1/32 and 1/16 inch thick. The other edge of the blade shall either be beveled in the same manner as the ruling edge or square, and it shall also be parallel to the ruling edge. Both edges of each blade shall be straight to within 0.01 inch per foot of length. The ends shall be square and at right angles to the edges. The rulers shall be assembled with the ruling edges outward.

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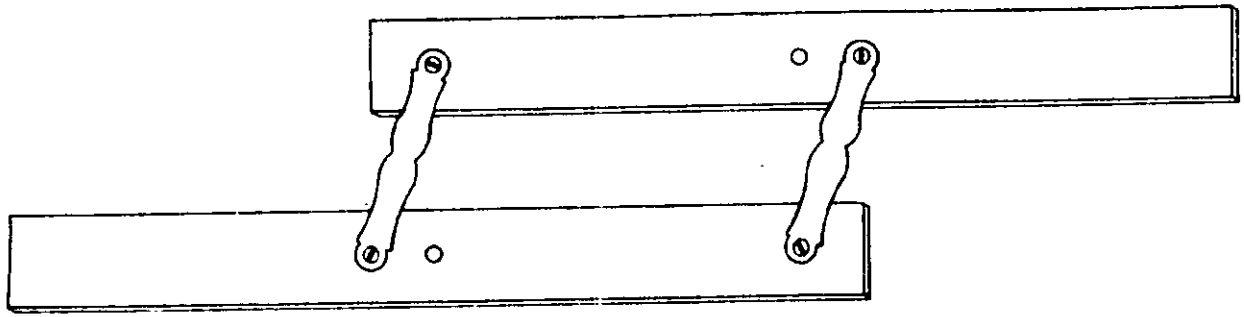


FIGURE 1. PARALLEL RULER

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3.3.2 Accuracy. When the instrument is in working position, the blades shall remain true to their common plane, and the ruling edge of both blades shall be parallel within 0.01 inch per foot of length at all positions.

3.4 Friction pads. One friction pad shall be securely attached to the underside of each blade near each end. On the 12-, 15-, 18-, and 24-inch rulers an additional pad shall be attached near the center of each blade. The pads shall be at least 1/2 inch in diameter and between 1/32 and 1/16 inch thick. Pads shall be of cork, rubber, or felt.

3.5 Bars. The bars shall be of metal or plastic. Nylon plastic, if used, shall conform to MIL-M-20693, Composition B. The bars shall be securely attached to the blade by metal rivets or screws in such a manner as to permit operation of the parallel ruler with minimum wear to the moving parts and to maintain parallelism of the ruling edges while in operation. The bars shall permit a minimum spread of 1-5/8 inches for the 6-inch size, and 3 inches for the 12-, 15-, 18-, and 24-inch sizes. The rivet heads or screws on the lower side of the blades shall be recessed above the surface. The 6-, 12-, 15-, and 18-inch rulers shall have two bars. The 24-inch ruler shall have three bars.

3.6 Knobs. A metal or plastic knob, not less than 1/4 inch nor more than 3/8 inch in height and not less than 1/4 inch nor more than 3/8 inch in diameter, shall be secured to each blade by a metal screw recessed into the underside of the blade.

3.7 Identification marking. The rulers shall be identified in accordance with MIL-STD-130.

3.8 Treatment and painting. Unless otherwise specified (see 6.2), the parallel rulers shall be painted black.

3.9 Class 1, wood blades. Each class 1 blade shall be of a continuous piece of hardwood. The wood shall be straight-grained and shall be free from knots, blemishes, and other defects. The wood shall be thoroughly seasoned and shall have a moisture content of not less than 6 percent nor more than 14 percent.

3.9.1 Finish. The wooden blades shall be given a protective coating of varnish. The blades shall be sanded before and after applying the varnish.

3.10 Class 2, plastic blades. Class 2 blades shall be of plastic conforming to L-P-398.

3.11 Workmanship. The parallel rulers shall be free from warpage, nicks, burrs, rough edges, scratches, and other imperfections which affect the appearance of which might impair the serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.4).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105.

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4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the following defects. Acceptable quality level (AQL) shall be 1.0 percent defective for major defects and 6.5 percent defective for minor defects.

Major

101. Material not as specified.
102. Protrusion below the surface or rivet heads or screws.
103. Dimensions not as specified.
104. Plastic cracked or chipped (plastic rulers only).
105. Edges not beveled.

Minor

201. Color and finish not as specified.
202. Friction pads, bars, and knobs not securely attached.
203. Identification marking incorrect, illegible, or missing.
204. Workmanship not as specified.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.3.3.1 through 4.3.3.4. AQL shall be 1.0 percent defective. Failure to pass any test shall be cause for rejection.

4.3.3.1 Ruling edge straightness. The ruling edge shall be placed against a precision flat, which shall be flat within 0.001 inch total. The separation between the ruling edge and the precision flat shall be measured at intervals not to exceed 3 inches with a feeler gage which has at least an accuracy of 0.001 inch. Repeat the test using the ruling edge of the opposite blade. Deviation from straightness of more than 0.01 inch per foot of length shall constitute failure of this test.

4.3.3.2 Parallel motion. Place a ruler upon a suitable drafting medium, and with a well-sharpened, hard-lead pencil draw a fine line along the entire length of one ruling edge. Spread the ruler to its maximum spread and draw another fine line along the same edge. Near each end of the first line, erect a perpendicular. Measure the distances from the base of the two perpendiculars to the points where they intersect the second line. Repeat the test using the ruling edge of the opposite blade. Lines not parallel within 0.01 inch per foot of length at all positions shall constitute failure of this test.

4.3.3.3 Parallel blade. Place a ruler upon a suitable drafting medium and with a well-sharpened, hard-lead pencil draw a fine line along the entire length of one ruling edge. Without moving the ruler, draw a fine line along the entire length of the ruling edge of the opposite blade. Near each end of the first line, erect a perpendicular. Measure the distances from the base of the two perpendiculars to the points where they intersect the second line. A difference of 0.01 inch per foot of ruled line in the distances measured along the perpendiculars shall constitute failure of this test.

4.3.3.4 Moisture content. The moisture content of wood rulers shall be determined with a resistance-type moisture meter in accordance with MIL-STD-1363. A moisture content of less than 6 percent or more than 14 percent shall constitute failure of this test.

4.4 Inspection of preparation for delivery.

4.4.1 Quality conformance inspection of pack.

4.4.1.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.4.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.4.1.3 Examination. Samples selected in accordance with 4.4.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106. Containers not as specified. Each incorrect container shall constitute one defect.
107. Rulers packed together not of like description.
108. Quantities packed together not as specified.
109. Weather-resistant fiberboard boxes not closed as specified.
110. Marking illegible, incomplete, or incorrect.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each ruler shall be placed in a close-fitting box conforming to PPP-B-566, variety 2, type, style, and class optional; PPP-B-665, class 2, style optional; or PPP-B-676, variety 2, type, style, and class optional.

5.1.2 Commercial. The rulers shall be preserved to afford protection against damage during shipment from the supplier to the initial destination in accordance with the requirements of Fed. Std. No. 356.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A. Rulers of like description shall be packed together in quantities as specified (see 6.2), in boxes conforming to PPP-B-601, Overseas type, Grade B, style optional. Strapping of the box is not required.

5.2.2 Level B. Rulers of like description in the quantities specified (see 6.2), shall be packed in boxes conforming to PPP-B-636, V3c, or equivalent. When specified (see 6.2), the boxes shall conform to PPP-B-636, class domestic, variety SW, Grade 275, or equivalent. Weather-resistant fiberboard boxes shall be closed in accordance with method V of the box specification appendix.

5.2.3 Commercial. Rulers of like description shall be packed together in quantities as specified (see 6.2), in containers that will assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Fed. Std. No. 356.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking specified by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military. In addition to any special marking required by the contract or order, marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.3 Commercial. Marking for commercial packaging shall be in accordance with Fed. Std. No. 356.

6. NOTES

6.1 Intended use. Parallel rulers are used in general drafting purposes and navigation to draw and lay out parallel lines.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type designation number (see 1.2.1).
(c) Color required, if other than as specified in 3.8.
(d) Preservation and packing required (see 5.1 and 5.2).
(e) Quantity to be packed (see 5.2.1, 5.2.2, and 5.2.3).
(f) When domestic fiberboard boxes are required (see 5.2.2).

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6.3 Figure 1 shows a type of parallel ruler which has been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of other rulers which conform to the specification.

MILITARY INTERESTS:

Custodians:

Army - ME
Air Force - 82

Review activity:

DSA - GS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
VA - DMS
COM - NBS

DOD Project No. 6675-0439

PREPARING ACTIVITY: ARMY - ME

: 1976 - 210-814/1114

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No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

GG-P-118C - Parallel Rulers, Folding Type

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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